

```
21 from keras.applications import InceptionResNetV2  ← 1
22 conv_base = InceptionResNetV2(weights='imagenet', include_top=False, input_shape=(150,150,3))  ← 2, 3, 4, 5
```

Figure : Importing the pre-trained model

2.1.Weights: The weights of the InceptionReNetV2 that was trained on the imagenet dataset can be obtained by using weights [imagenet] as a parameter.

2.2.include_top: This parameter tells Keras whether to download the fully connected layers of the pre-trained model that do the final classification. It means that after the convolution layers extract basic features such as edges or lines from the input images, the deeper fully connected layer then classifies them.

Since we are going to build only a two class (dogs and cats) classifier on our model, we will specify the parameter to be include_top [False].

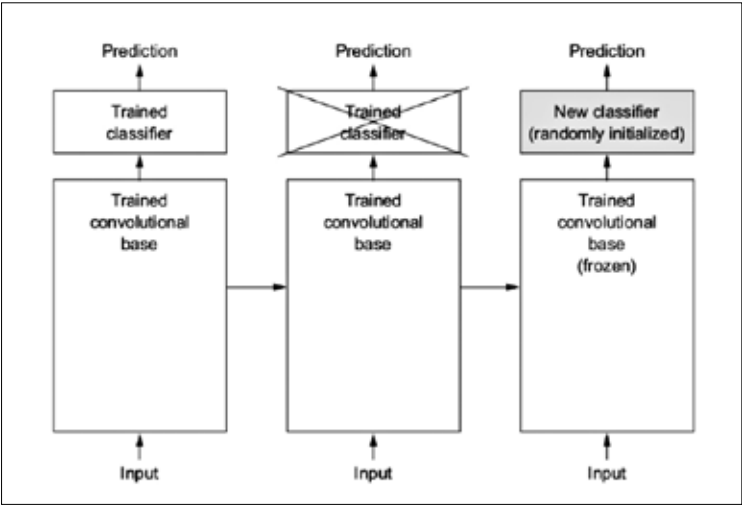


Figure : Top Layers of the classifier

3. Now, we download the model. We can call the .summary() function on the InceptionResNetV2 model to see its architecture and number of parameters. There are 54 million plus parameters. Therefore, we cannot train it from scratch. Due to the power of Transfer learning, we are re-using it without training.